

A manifest of docs/reports

Sixty-four independent mathematical research packages,
classified by subject

Vladimir Reshetnikov

19 September 2026

1 Scope and conventions

This document catalogues the sixty-four research packages stored under **docs/reports**. Each package was delivered as a single ZIP archive; the archives have been unpacked, one directory per package, and grouped into the subject hierarchy described in Section 2. The archives themselves are no longer kept in the repository.

Every package is self-contained and follows the same shape: a typeset article (***.pdf**) with its complete **L^AT_EX** source, a **README.md** summarising the result, usually a **code/** directory with exact-arithmetic verification scripts, a **data/** or **results/** directory holding the recorded output of those scripts, and often **STATUS.md**, **sources.md** or **notes/** files auditing the sources consulted and the precise limits of the claim.

Two cautions apply uniformly and are therefore not repeated in the individual entries. First, these are AI-assisted research drafts: none has been independently refereed, and none has been checked in a proof assistant. Second, the descriptions below record *what each report sets out to settle and what it claims to prove*. Compiling this manifest involved reading each package’s statement of its problem and result; it did not involve checking any proof. Where a report itself flags a limitation — a prior disproof, an imported theorem, a correction to the source, or an exception left open — that is noted, because it changes what the report is.

The naming of the directories is editorial. Several distinct reports attack the same published problem, and some arrived under colliding archive names, so each directory has been given a name describing its own angle on the problem. The original archive name is recorded with every entry.

2 The directory tree

Category	Subcategory	Reports
ordinals-and-order-types	friendly-order-types	4
	wqo-powersets-and-statutes	4
	transfinite-words	3
	ordinal-arithmetic	2
	games-on-ordinals	2
hankel-determinants	somos-and-elliptic	3
	catalan-and-ballot	2
	arithmetic-hankel	2
	growth-and-runs	2
congruences-and-valuations	supercongruences	3
	iterated-series	2
	(top level)	3
tetration-and-digit-stabilization	—	7
log-concavity-and-unimodality	—	7
graph-theory	—	3
enumerative-combinatorics	—	8
generating-functions-and-asymptotics	binary-product-coefficient-counts	4
	(top level)	3
Total		64

More than a quarter of the collection consists of counterexamples and corrections rather than proofs. Entries 14, 15, 29, 32, 33, 38, 39, 40, 41, 43, 48, 52, 58, 59, 60, 62 and 64 refute, or replace with a proved statement, an assertion published or catalogued elsewhere; entry 34 disproves one of the three conjectures it treats while proving the other two. Several reports whose main outcome is a proof still correct the statement they prove: entries 17, 19 and 24 record a sign error, a missing one-unit shift and a numbering ambiguity in their sources, and entry 44 refines its target after finding that the original conjecture had already been disproved by someone else.

3 Ordinals and order types

3.1 Friendly order types

Four independent reports on Isa Vialard’s *friendly order type* $f(P)$, the rank of the tree of open-ended bad sequences in which every chosen point has an incomparable partner in the current residual. All four prove the same finite formula; they differ in what they build on top of it.

1. Friendly Order Type and Incomparability Components

[ordinals-and-order-types/friendly-order-types/incomparability-components-and-forests/article.pdf](#)
 from `friendly_order_type_research.zip`

Proves $f(P) = |P| - c(\text{Inc}(P))$ for a finite poset P , where c counts connected components of the incomparability graph (isolated vertices included). With Vialard’s width-transfer theorem this gives ordinal width $\omega^{f(P)}$ for the Dershowitz–Manna multiset ordering. Also classifies all maximum friendly subsets, builds spanning-forest certificates, and gives Cartesian-product, lexicographic-substitution and well-ordered-sum formulas with an enumerative generating function.

2. Friendly Order Types Through Incomparability Components

ordinals-and-order-types/friendly-order-types/
incomparability-components-and-infinite-tails/friendly_order_types.pdf
from friendly_order_type_research (1).zip

The same component formula, together with an exact friendly rank for connected infinite well-partial-orders: writing $o(C) = \lambda + n$ with λ the limit part and $T(C)$ the finite set of points with finite principal upsets. Includes an exhaustive finite verification and a proof audit.

3. Friendly Order Type Through Incomparability Components

ordinals-and-order-types/friendly-order-types/
matroid-rank-and-ordinal-sums/article.pdf
from friendly_order_type_research (3).zip

Identifies $|P| - c(P)$ with the graphic-matroid rank and the incidence-matrix rank of the incomparability graph. Adds a substitution formula, a finite Cartesian-product formula, and additivity over arbitrary well-ordered ordinal sums; evaluates every well-partial-order whose incomparability components are finite, and gives countable counterexamples delimiting stronger infinite extrapolations.

4. Friendly Order Types of Finite Posets and Ordinal Products

ordinals-and-order-types/friendly-order-types/
cartesian-products-of-ordinals/friendly_order_types.pdf
from friendly_order_types_research.zip

Beyond the component formula: a Cartesian product with at least two non-singleton factors has exactly one non-endpoint incomparability component; the tuple (o, h, w, f) of the factors does *not* determine $f(A \times B)$ (explicit four-point D, E with equal tuples and $f(D \times 2) = 5$, $f(E \times 2) = 6$); and a complete classification of $f(\alpha_1 \times \cdots \times \alpha_r \times B)$ as a natural product, with an exception when every α_i is a successor and B has a greatest element.

3.2 Well-quasi-ordered powersets and statures

Four reports on the open questions in Section 6 of Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard, *Measuring well-quasi-ordered finitary powersets* (arXiv:2312.14587v2), and on the corresponding topological invariant.

5. Cofinal strata and ordinal absorption

ordinals-and-order-types/wqo-powersets-and-statures/
cofinal-strata-of-finitary-powersets/article.pdf
from cofinal_strata_ordinal_powersets.zip

For $S_\alpha(Q)$, a finite poset Q with each point replaced by a chain $\alpha = \omega^\rho$, proves $o(K(S_\alpha(Q))) = \sum_k \omega^{\rho \otimes k} c_k(Q)$ and $h(K(S_\alpha(Q))) = \alpha \cdot \text{height}(M(Q))$, where c_k counts size- k maximal antichains not dominated by a larger one. The maximal-order-type theorem allows nonuniform pure well-partial-order fibers and is proved by a finite cofinal-stratification argument.

6. Intrinsic ordinal approximations for finitary powersets

ordinals-and-order-types/wqo-powersets-and-statures/
intrinsic-ordinal-approximations/article.pdf
from intrinsic_ordinal_approximations.zip

Studies the intrinsic invariant $a_o(A) = \sup\{o(B) + 1 : \text{Idl}(B) \hookrightarrow A\}$. Proves the universal bound $a_o(A) \leq h(P_f(A))$ with equality exactly when every finitely generated downset of A is an algebraic depo, gives the closed formula $h(P_f(\prod_i \alpha_i)) = \sup(\bigotimes_i (1 + a_i) + 1)$, and shows the universal identity fails on $\lambda + F$.

7. Ordinal Order Maps and Finitary Powersets of Ordinal-Finite Grids

ordinals-and-order-types/wqo-powersets-and-statures/
ordinal-order-maps-and-grids/ordinal_order_maps.pdf

from ordinal_order_maps_research.zip

Exact maximal order types and heights for finite Hoare powersets of $\beta \times Q$ with Q a finite poset: for $\beta = \omega^{\gamma_0} + \dots + \omega^{\gamma_{m-1}}$, $o(\text{Mon}(Q, \beta)) = \bigoplus_{s \text{ isotone}} \omega^{\bigoplus_q \gamma_{s(q)}}$.

8. Sharp ordinal bounds for Hoare powerspaces

ordinals-and-order-types/wqo-powersets-and-statures/
hoare-powerspace-statures/article.pdf

from hoare_statures_research.zip

For all Noetherian spaces, $1 + \|X\| \leq \|HX\| \leq 2^{\|X\|}$, with witnesses attaining both endpoints at every ordinal and T_1 upper witnesses; the older ω^α bound is attainable at infinite input stature α exactly when $\omega \cdot \alpha = \alpha$; the complete spectrum at $\omega + n$ is $\omega \cdot j(U) + (j(Q) - j(U))$. Extends the known well-quasi-order results of Abriola et al. to arbitrary Noetherian spaces.

3.3 Transfinite words

Three reports on Harry Altman, *Bounding finite-image sequences of length ω^k* (arXiv:2409.03199v2), Example 3.15: is the bound ω^{ω^4} sharp for words of length $< \omega^2$ over the two-element antichain?

9. Exact maximal order types of finite-alphabet transfinite words

ordinals-and-order-types/transfinite-words/
finite-alphabet-transfinite-words/article.pdf

from finite_alphabet_transfinite_words.zip

Proves $o(W_k(P)) = \omega^{\omega^{N_k(P)-1}}$ for words of ordinal length $< \omega^k$ over a finite poset P under subsequence embedding, where N_k counts indecomposable classes of a canonical finite atom poset $I_k(P)$, with $N_1(P) = |P|$ and $N_{k+1}(P) = |P| + J(I_k(P)) - 1$. For $k = 2$ the binary antichain gives ω^{ω^4} — the tightness Altman leaves open — and the binary chain gives ω^{ω^3} .

10. Exact maximal order types below omega squared

ordinals-and-order-types/transfinite-words/order-types-below-omega-squared/article.pdf
from omega2_order_types_proposed_solution.zip

The $k = 2$ case in its own right: $o(s_{\omega^2}^F(P)) = \omega^{\omega^{n+j(P)-2}}$ for $n = |P| \geq 2$, with $j(P)$ the number of downsets including the empty one. Handles the empty and singleton alphabets as genuine exceptions and classifies arbitrary selected families of periodic tail types.

11. Guarded periodic blocks and maximal order types below omega squared

ordinals-and-order-types/transfinite-words/guarded-periodic-blocks/article.pdf
from ordinal_words_research.zip

The same formula $o(s_{\omega^2}(X)) = \omega^{\omega^{n+|J(X)|-2}}$, obtained by a guarded-periodic-block construction, with symbolic algorithms and executed tests; the one-letter alphabet (type ω^2) is flagged as outside the formula's range.

3.4 Ordinal arithmetic

Two independent proposed solutions to Problem 6.2 of Paolo Lipparini, *Monotone infinitary operations on ordinals* (arXiv:2505.00424v2): define and study the least weakly monotone, e -special strictly increasing operation on countable sequences of ordinals.

12. An Explicit Formula for the Minimal e -Monotone Infinitary Ordinal Sum

ordinals-and-order-types/ordinal-arithmetic/lipparini-minimal-infinitary-sum/article/
explicit_ordinal_sum.pdf

from lipparini_problem_6_2_proposed_solution.zip

An explicit formula for the least admissible operation, proved by an arithmetic admissibility upper bound and a rank-theoretic minimality lower bound, using an explicit countable-multiset profile order, its rank, absorption and correction rules, bounded-poset heights and finite-cap ranks. Imports Lipparini's formula and minimality theorem for the weaker operation S .

13. An Explicit Formula for Lipparini's Minimal Infinitary Ordinal Operation

ordinals-and-order-types/ordinal-arithmetic/
 lipparini-minimal-operation-formula/article.pdf
 from ordinal_operation_research.zip

A three-case finite ordinal-arithmetic formula with proofs of admissibility and pointwise minimality. Relative to Lipparini’s already evaluated S , the correction is zero, a finite increment, or one *natural* summand ω — ordinary addition of ω is not a substitute.

3.5 Games on ordinals

14. A counterexample to winner stability in ordinal Chomp

ordinals-and-order-types/games-on-ordinals/
 ordinal-chomp-winner-stability/ordinal_chomp_counterexample.pdf
 from ordinal_chomp_counterexample.zip

For the numerical semigroup $S = \langle 4, 6, 9 \rangle$, Chomp on S is a second-player win while Chomp on the ordinal square $S^2 = \{\omega a + b : a, b \in S\}$ is a first-player win — the opening $\omega \cdot 4 + 4$ leaves Grundy value zero, and the same move wins at every exponent ≥ 2 , so $\text{ch}(4, 6, 9) = 2$. This refutes Conjecture 4.1 of arXiv:2504.07317v1 and answers Question 4.6 of v3 negatively, already for natural-number generators.

15. Open-query games on ordinal spaces

ordinals-and-order-types/games-on-ordinals/open-query-membership-games/article.pdf
 from ordinal_membership_games.zip

A negative answer to Problem 1.3 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3): two copies of $[0, \omega]$ each have set-membership number 1, but their topological sum has value 2. The general theorem computes the invariant for every nonempty Hausdorff space of finite Cantor–Bendixson height h as $\lceil \log_2(h + 1) \rceil$ when the last nonempty derivative is a singleton and $\lceil \log_2(h + 2) \rceil$ otherwise.

4 Hankel determinants and continued fractions

4.1 Somos sequences and elliptic curves

16. A proof of Barry’s Hankel–Somos-6 conjecture

hankel-determinants/somos-and-elliptic/barry-somos-6-hankel-recurrence/article.pdf
 from barry_somos6_research.zip

Conjecture 7 of Paul Barry, arXiv:2211.12637: for the Hankel determinants H_N of a four-parameter Jacobi-type generating function, $H_N H_{N-6} = \alpha H_{N-1} H_{N-5} + \gamma H_{N-3}^2$ with $\alpha = t^2(r + 2)^2$ and $\gamma = t^3((s + t)(r + s + t + 2)^2 + t(r + 2)^3)$. Also proves a homogeneous four-parameter form and a seven-parameter polynomial Hankel theorem making exceptional specialisations rigorous; the verifier never divides by a determinant.

17. Finite Hankel Certificates for Three Somos-8 Conjectures

hankel-determinants/somos-and-elliptic/somos-8-hankel-certificates/article.pdf
 from somos_hankel_proofs.zip

Conjectures 11, 12 and 13 of the same Barry paper, by importing Hone’s genus-two Hankel theorem (arXiv:1907.05204v3, Thm. 5.4) and supplying a binomial translation, a polynomial-minor specialisation valid on the degenerate locus, and a rank-bound lemma that turns four symbolic rows into an all-index identity. Identifies a sign error in the source’s Catalan expression for Conjecture 13.

18. Elliptic addition and Jacobi continued fractions

hankel-determinants/somos-and-elliptic/
 elliptic-addition-and-jacobi-fractions/article.pdf

from `elliptic_continued_fraction_research.zip`

A consistently indexed, non-torsion version of Barry's 2023 Conjecture 4, with the coordinate formulas of Conjecture 3: every continued-fraction tail is given explicitly and removing one layer is shown to be elliptic translation. Hypotheses: characteristic zero, $Y^2 + aXY + bY = X^3 + cX^2 + dX$ nonsingular, $b \neq 0$, and $P = (0, 0)$ of infinite order; order-three and order-four examples show the regular fraction genuinely needs the last one.

4.2 Catalan and ballot moments

19. Shifted Catalan Hankel Polynomials

[hankel-determinants/catalan-and-ballot/shifted-catalan-hankel-polynomials/article.pdf](#)

from `catalan_hankel_conjecture_solution.zip`

For $D_N^{(m)}(a, b) = \det(aC_{i+j+m} + bC_{i+j+m+1})$, proves the denominator-exponent assertion on page 4 of Barry, arXiv:2011.10827v1, and shows the separately printed Conjecture 2 needs an explicit one-unit shift correction. The minimal characteristic polynomial is $(X^2 - (a+2b)X + b^2)^{m(m-1)/2+1}$, so the minimal order is $m(m-1) + 2$; all exceptional reduced denominators, the exact coefficient formula, numerator degree and reciprocal symmetry are supplied.

20. Product formulas for ballot-polynomial Hankel determinants

[hankel-determinants/catalan-and-ballot/](#)

[ballot-polynomial-hankel-determinants/article.pdf](#)

from `Cigler_Hankel_Conjectures_13_15.zip`

Conjectures 13, 14 and 15 in Section 5 of Johann Cigler, arXiv:2111.14492v3. An explicit coefficient product gives the sign and exact support of the normalised shift- k Hankel determinant of the ballot moments and positivity of every residual coefficient for $n \geq 1$; the same products give the conjectured generating-function denominators, exact numerator degrees and reciprocal numerator identities. Adds strict log-concavity of the residual coefficients and their limiting binomial profile.

4.3 Arithmetic Hankel determinants

Two independent reports on the numerator formula recorded as a conjecture in OEIS A122251, with A122249 as its binary special case.

21. A Proof of an OEIS Hankel-Numerator Conjecture

[hankel-determinants/arithmetic-hankel/a122251-numerator-formula/report.pdf](#)

from `hankel_numerator_research.zip`

For $H_n(m, a) = \det(1/(a+m(i+j)))_{0 \leq i, j \leq n}$, the reduced numerator is $\prod_{p|m} p^{n(n+1)v_p(m)+2 \sum_{k \leq n} v_p(k!)}$: independent of the coprime offset a , a perfect square, and supported on the primes dividing m . Also gives exact denominator valuations, the gcd of denominators over all coprime a , a constructive congruence witness, the noncoprime-parameter formula, recurrences and a natural-boundary proof.

22. Prime-supported numerators of arithmetic Hankel determinants

[hankel-determinants/arithmetic-hankel/a122251-prime-exclusion-proof/report.pdf](#)

from `hankel_oeis_conjecture_proof.zip`

The same formula in the size- m indexing, $\prod_{\ell|d} \ell^{v_\ell(d)m(m-1)+2 \sum_{j < m} v_\ell(j!)}$, with the exclusion of every prime not dividing d carried out as a separate step — the point being that determining the p -adic valuation alone would not suffice. Every reduced denominator valuation is determined as well.

4.4 Growth laws and run statistics

23. A proof of the Apéry Hankel growth conjecture

[hankel-determinants/growth-and-runs/](#)

[apery-hankel-determinant-growth/apery_hankel_growth.pdf](#)

from [apery_hankel_research.zip](#)

For $A_k = \sum_j \binom{k}{j}^2 \binom{k+j}{j}^2$ and $D_n = \det(A_{i+j})$, proves $\log D_n = n(n+1) \log \frac{17+12\sqrt{2}}{4} + O(n)$ — the limit conjectured in the September 2026 revision of OEIS A228143 — using Edgar’s moment-density theorem. Also an explicit law for $\det(A_{m(i+j)+r_n})$ with $r_n/n \rightarrow \rho$, and non-D-finiteness of the formal generating series of D_n .

24. Binary Runs in Rueppel Hankel Determinants

[hankel-determinants/growth-and-runs/rueppel-binary-run-determinants/rueppel_hankel.pdf](#)

from [rueppel_hankel_research.zip](#)

For the Rueppel series $r(x) = \sum_{j \geq 0} x^{2^j-1}$, the polynomial identity $H_n(tx + 1/r(x^2)) = (-1)^{1+\varepsilon+\lfloor R(n)/2 \rfloor} (\varepsilon + (R(n) - \varepsilon)t^2)$, where $R(n)$ counts runs of equal bits in the binary expansion of n and ε is the low bit of the Gray code $n \oplus \lfloor n/2 \rfloor$. Settles statements 8, 10, 13 and 23 of Barry’s 2021 paper (not the identically numbered ones elsewhere), with a correction.

5 Congruences and p -adic valuations

5.1 Supercongruences

25. A cubic supercongruence for OEIS A348410

[congruences-and-valuations/supercongruences/a348410-cubic-supercongruence/article.pdf](#)

from [A348410_Supercongruence_Research.zip](#)

For $A_{u,v}(n) = [x^n](1-x)^{-un}(1+x)^{-vn}$ proves $v_p(A_{u,v}(N) - A_{u,v}(N/p)) \geq 3v_p(N) - \varepsilon_p$ with $\varepsilon_3 = 1$ and $\varepsilon_p = 0$ for $p \geq 5$, for every odd prime p and every N divisible by p — no coprimality condition on the remaining multiplier. Taking $(u, v) = (2, 1)$ gives Peter Bala’s A348410 supercongruence. Appendix A audits arXiv:2104.10754v1, whose prime-by-prime framing conclusion and weighted harmonic assertion fail on its own displayed examples.

26. Small-prime congruences for A348410

[congruences-and-valuations/supercongruences/a348410-small-prime-congruences/article.pdf](#)

from [a348410_supercongruence.zip](#)

For $a(n) = [x^n]((1-x)(1-x^2))^{-n}$ proves $v_p(a(mp^r) - a(mp^{r-1})) \geq 3r - v_p(12)$ for every prime and all $m, r \geq 1$, hence $(12/n^3) \sum_{d|n} \mu(d)a(n/d) \in \mathbb{Z}_{\geq 0}$ with 12 optimal. The principal extension is an exact binary leading-defect formula for the family $A_{\alpha,\beta}$, valid modulo 2^{3r-2} .

27. Signed binomial supercongruences and OEIS A352373

[congruences-and-valuations/supercongruences/](#)

[a352373-signed-binomial-supercongruences/signed_binomial_supercongruence.pdf](#)

from [OEIS_A352373_supercongruence.zip](#)

For $a(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$ proves $a(mp^h) \equiv a(mp^{h-1}) \pmod{p^{3h}}$ for $p \geq 5$ and $a(m3^h) \equiv a(m3^{h-1}) \pmod{3^{3h-1}}$, with m not required coprime to p . Covers A352373, A348410, A351856 and A351857 as specialisations; the only non-elementary import is the classical Jacobsthal binomial congruence.

5.2 Congruences for iterated and compositional series

28. Arithmetic of a Self-Iterating Generating Function

[congruences-and-valuations/iterated-series/a396807-self-iterating-series/article.pdf](#)

from [A396807_congruences_and_prime_power_lifts.zip](#)

Both conjectures in OEIS A396807 for $A(x) = x + A^{[5]}(x)A^{[6]}(x)$ (functional iteration): $A(x) \equiv x/(1-x)$ and $A^{[k]}(x) \equiv x/(1-kx)$ modulo 10 for every integer k , negative iterates included. Adds a sharp modulus theorem for $F = x + cF^{[r]}F^{[s]}$, rational lifts at every selected prime power, and minimal period 46860 for A396807 modulo 100 starting at index 1.

29. Prime-power congruences for compositional tree series

[congruences-and-valuations/iterated-series/compositional-tree-series-congruences/article.pdf](#)
from [compositional_tree_congruences.zip](#)

For $A_l(x) = x \exp(A_l^{ol})$, shows the OEIS A396805 claim $a_5(n) \equiv n \pmod{5}$ is *false*: first failure $n = 23$, with exception set $\{20j + 3, 20j + 4 : j \geq 1\}$ and residue 2 throughout. Proves eleven other conjectural assertions in A396803–A396806 via a general reduction of coefficients mod p^α to typed rooted trees on fewer than p^α labels, yielding eventual periods, rational residue generating functions, and an exact classification of $a_l(n) \equiv n \pmod{p}$.

5.3 Valuations, divisibility and periodicity

30. Binary Carries and the Divisibility of Two-Stack-Sortable Permutation Numbers

[congruences-and-valuations/a000139-binary-carries-parity/research.pdf](#)
from [a000139_carries_research.zip](#)

Proves Peter Bala’s parity conjecture for OEIS A000139, $a(n) = 2(3n)!/((n+1)!(2n+1)!)$: $a(n)$ is odd exactly when n is odd and its binary expansion contains no 11. The full valuation is $v_2(a(n)) = s(n) + s(n+1) - s(3n)$ in binary digit sums. Two independent parity proofs, plus exact valuation distributions, recurrences, asymptotics, extremal formulas and limit laws.

31. Polynomial divisibility of height-one factorial ratios

[congruences-and-valuations/factorial-ratio-polynomial-divisibility/article.pdf](#)
from [bala_factorial_divisibility_research.zip](#)

A general rational-root criterion for uniform polynomial divisibility of balanced integral factorial ratios, applied to Peter Bala’s uniform-product conjectures for OEIS A211417 and A295431: explicit multipliers for every product length, with the least multipliers certified in eight particular examples. Includes an atlas of the 52 sporadic height-one ratios with recurrences, generating functions and asymptotic expansions.

32. When the Secant Numbers Become Periodic

[congruences-and-valuations/secant-number-periodicity/article.pdf](#)
from [secant_periodicity_research.zip](#)

Refutes the OEIS A000364 conjecture that the secant numbers from $n = 1$ are purely periodic modulo every m : at $m = 27$, $a(1) \equiv 1$ but $a(19) \equiv 10$. The exact replacement is that pure periodicity from index one holds if and only if no odd prime cube divides m ; the least eventual period always divides $\varphi(m)$. Determines the least eventual period and exact onset for every modulus, with densities and a certified mean onset.

6 Tetration and digit stabilization

33. A Counterexample to a Decimal-Length Bound for Tetration Stabilization

[tetration-and-digit-stabilization/congruence-speed-decimal-length-bound/article.pdf](#)
from [tetration_counterexample.zip](#)

An explicit 2544-digit base $A = 1 + qu$ ($q = 5^{2547}$) for which $\nu_{10}(T_{n+1} - T_n) = \min(1 + 2548n, 2547(n+1))$ at every height, so the congruence speed at height $\text{len}(A) + 2$ is 2548 while the permanent value is 2547. This contradicts the decimal-length form of Conjecture 1 of Marco Ripà, *The congruence speed formula* (NNTDM 27(4), 2021). The report audits the later Ripà–Onnis confirmation: their weaker valuation bound is compatible with the counterexample, the decimal-length bound is not.

34. Exactly One Digit per Height

[tetration-and-digit-stabilization/one-stable-digit-per-height/article.pdf](#)

from tetration_exact_digits.zip

The three conjectural comments in OEIS A324017: the first two are proved and the third is disproved for every nontrivial base in its stated domain. For even $q \geq 4$ and $a = q - 1$, the central theorem is $T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}$, so each added height contributes exactly one new stable radix- q digit. Smallest counterexample to the third: $3 \uparrow\uparrow 2 \equiv 27$ and $3 \uparrow\uparrow 3 \equiv 59 \pmod{64}$. Adds exact prime-adic distances, a factorization-free digit-lifting algorithm and higher Knuth arrow ranks.

35. Tetration, perfect powers, and frozen decimal digits

[tetration-and-digit-stabilization/perfect-power-minima-for-frozen-digits/article.pdf](#)
from tetration_perfect_power_minima.zip

Determines $\rho(n)$, the smallest n th perfect power $x^n > 1$ whose eventual number of newly frozen terminal decimal digits is exactly n : $\rho(1) = 2$, $\rho(2) = 49$, and $\rho(n) = B(s)^n$ for $n \geq 3$ with $s = n - v_2(n)$ and $B(s) \in \{2^s \pm 1, 3 \cdot 2^s \pm 1\}$ according to $s \pmod{4}$. All other roots are ruled out by a deterministic quadratic size barrier plus small exact certificates.

36. Phase-shift classification for integer tetration

[tetration-and-digit-stabilization/phase-shift-classification/tetration_phase_classification.pdf](#)
from tetration_phase_classification.zip

Proves the 21-word phase-shift conjecture in OEIS A376842, the 14-word even-base restriction from the appendix of Ripà's 2025 paper, and the 28-word prescribed-anchor conjecture in OEIS A376446. The mechanism compares the 2-adic and 5-adic valuations of $D_b = T_{b+1} - T_b$ at the first permanently constant-speed height: strict 2-adic dominance gives phase 5, strict 5-adic dominance a multiplicative orbit in $\mathbb{F}_5^\times$, and balanced valuations exactly four further anchored words.

37. A Sharp Convergence Theorem for the q -Newton Series of Subcritical Tetration

[tetration-and-digit-stabilization/q-newton-series-convergence/article/tetration_qnewton.pdf](#)
from tetration_qnewton_research.zip

Proves convergence of the explicit q -binomial interpolation series of MathOverflow question 259278 for every real base $1 < a < e^{1/e}$, with $\log a = qe^{-q}$: the grouped series converges absolutely and locally uniformly for $\operatorname{Re} z > -2$, converges conditionally on $\operatorname{Re} z = -2$ except at $z = -2 + 2\pi ik/(-\log q)$, and diverges for $\operatorname{Re} z < -2$. Adds strict complete monotonicity and uniqueness under an eventual log-convexity condition.

38. Delayed stabilization of power towers

[tetration-and-digit-stabilization/delayed-stabilization-of-towers/article.pdf](#)
from tetration_stabilization_research.zip

Refutes the claim that every prime base ending in 9 reaches its permanent decimal congruence speed by height 2 (Ripà, Conjecture 2, journal p. 57; arXiv:2208.02622v1, Conjecture 4.1): 2749 is the smallest prime counterexample, with first permanent-speed height $H(2749) = 3$. Proves exact formulas for $C_a(h)$, $V(a)$ and $H(a)$ in terms of $s = v_2(a - 1)$, $t = v_2(a + 1)$, $u = s + t - 1$, $e = v_5(a + 1)$, and shows there are infinitely many primes with each prescribed onset $H \geq 3$.

39. Delayed digit stabilization: a counterexample and sharp bounds

[tetration-and-digit-stabilization/iterated-power-digit-stabilization/delayed_digit_stabilization.pdf](#)
from digit_stabilization_research.zip

For $S_a(b) = v_{10}(a^{10^{b+1}} - a^{10^b})$ and $D_a(b) = S_a(b) - S_a(b - 1)$, answers negatively the question whether $D_a(b) = 1$ for all $b \geq 3$: the smallest counterexample is $a = 3 \cdot 2^{99} + 1$, with $S_a(2) = 100$, $S_a(3) = 102$, $D_a(3) = 2$. Proves minimality, an exact onset formula, the smallest counterexample at every stage, the natural density of exceptions, and arbitrary-radix extensions. The article is explicit that a^{10^b} is not genuine tetration and derives the tower formulas separately.

7 Log-concavity, unimodality and positivity

40. Single curves do not force log-concavity

[log-concavity-and-unimodality/cluster-variable-log-concavity-counterexample/article.pdf](#)
from [annular_single_lamination_counterexample.zip](#)

An unpunctured annulus with one marked point on each boundary and a single unit-weight simple lamination curve gives, after mutations 2, 1, 2 with the two mutable variables specialised to 1, the polynomial $1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6$; the consecutive coefficients 3, 2, 4 form a strict valley, refuting Conjecture 1 of arXiv:2508.04396v3 and conflicting with its general unpunctured-surface unimodality statement. The geometric realisation is proved, not merely the matrix calculation, and a fixed seed gives infinitely many such examples.

41. Cycles refute infinite log-concavity for chromatic coefficients

[log-concavity-and-unimodality/infinite-log-concavity-of-cycles/article.pdf](#)
from [chromatic_cycle_logconcavity.zip](#)

For the cycle C_n with absolute chromatic coefficients under the iterated operator $L(a)_k = a_k^2 - a_{k-1}a_{k+1}$: infinitely log-concave for $3 \leq n \leq 11$; first failure at depth 5 for $n = 12$, depth 4 for $13 \leq n \leq 16$, and depth 3 for $n \geq 17$. So C_{12} is the smallest cycle counterexample (no minimum over all graphs is claimed), with an explicit third-iterate witness polynomial valid for all $n \geq 3$.

42. Oscillating Log-Concavity in Geometric Generating Functions

[log-concavity-and-unimodality/oscillating-log-concavity-geometric-transform/article.pdf](#)
from [oscillating_log_concavity_research.zip](#)

Challenges 2 and 3 in Section 3 of Heim and Neuhauser, arXiv:2302.13327v1, for the *geometric* transform $1/(1 - uG(z))$. For divisor-sum weights, every real exponent and every $u > 0$ give infinitely many strict log-concavity and log-convexity indices with eventually bounded gaps; likewise for integer power weights with exponent ≥ 2 , the exponents 0 and 1 being explicitly exceptional. For square weights at $u = 1$, every four consecutive indices carry both signs (sharp), each with density 1/2 in every arithmetic progression. A subdominant-pole theorem is the common mechanism.

43. A counterexample and a sharp spacing-three criterion for products of q -integers

[log-concavity-and-unimodality/q-integer-product-unimodality/report.pdf](#)
from [q_unimodality_attack.zip](#)

The necessity half of Conjecture 5.4 of Connelly, Ito, Martinez, Shevchenko and Yang (arXiv:2605.12822v1) is false: $(1 + q)^6(1 + q^3)$ is symmetric and weakly unimodal although no a_i is divisible by 3 and $b = 2$ exceeds the asserted bound, and the example lies inside the source's claimed range. The report also proves the proposed sufficient condition for every $r \geq 2$, a necessary degree bound in the absence of an r -divisible ordinary factor, and a complete classification of spacings two and three.

44. Sharp size thresholds for failures of log-concavity in independence systems

[log-concavity-and-unimodality/independence-system-thresholds/article.pdf](#)
from [sharp_log_concavity_thresholds.zip](#)

For each rank $r \geq 4$, determines the smallest ambient ground-set size admitting failure of ordinary, ordered and ultra log-concavity, and shows the three thresholds coincide across all independence systems, those satisfying Xie–Xu's augmentation bound, and those with additive augmentation parameter 3 (values 7/6/5, 7/7/6, 8/7/7, and $r + 1$ throughout for $r \geq 7$). The initial target, Xie–Xu Conjecture 4.8 (arXiv:2408.09152v1), had already been disproved by Alex Chengyu Li in September 2026; the report states this and refines the target to the rankwise minimum-size question.

45. A sharp order-five threshold for higher-order Stirling subset log-concavity

[log-concavity-and-unimodality/higher-order-stirling-rows/article.pdf](#)

from `stirling_log_concavity.zip`

For $S_{n,k}^{(r)}$, the number of partitions of $n + (r - 1)k$ labels into k blocks of size $\geq r$, proves that all rows are log-concave if and only if $1 \leq r \leq 5$, with strict inequalities in the interior — the log-concavity assertion of Deb–Sokal Conjecture 1.4(c) (arXiv:2507.18959v1). Also proves the cycle cases $r = 3, 4$; the fifth-order *cycle* conjecture is explicitly left unresolved, and the included operator counterexample is flagged as not being a counterexample to it.

46. A sum-of-squares proof of Ulas’s real-zero conjecture

`log-concavity-and-unimodality/stern-polynomial-positivity/stern_conjecture_note.pdf`
from `stern_conjecture_research.zip`

Conjecture 4.4 of Maciej Ulas, arXiv:1909.10844v1, for the Stern polynomials indexed by $a_n = (4^n - 1)/3$ and $h_n = 2(4^n - 1)(2 \cdot 4^n + 1)/3 + 1$: for every $n \geq 0$ and every real t , $W_n(t) = (A_{n+1}(t) - tA_n(t)/2)^2 + 3t^2 A_n(t)^2/4 > 0$. The adjacent-polynomial quadratic identity behind it was already in Ulas’s paper and is re-derived, not claimed as new.

8 Graph theory

47. Mutual-visibility sets in Andrásfai graphs

`graph-theory/andrasfai-mutual-visibility/article.pdf`
from `andrasfai_visibility_research.zip`

Proves the generating function attributed to Andrew Howroyd in OEIS A391632, $\sum a(n)x^n = x(4 - 19x + 33x^2 - 40x^3 + 8x^4)/(1 - 10x + 29x^2 - 32x^3 + 8x^4)$, by an exact weight-preserving bijection between mutual-visibility sets of the n -Andrásfai graph and anchored closed walks of length $3n - 1$ in a four-state labelled graph. Adds the full bivariate generating function, an exact binomial formula for every cardinality, maximum-set classification, eventual polynomiality at each fixed deficit, exponential asymptotics, a central limit theorem, and linear-time recognition and exact uniform sampling.

48. A bicyclic 30-vertex graph with domination root -4

`graph-theory/domination-root-minus-four-order-30/domination_root_30.pdf`
from `domination_root_30_research.zip`

An explicit connected planar graph on 30 vertices and 31 edges, treewidth 2 and domination number 10, with $D(G, x) = x^{10}(x + 4)Q_{19}(x)$ — a negative answer to question 2 of Section 4 in Alikhani and Griswold, arXiv:2608.00109v1, which asks whether 33 is the minimum order. Their 33-vertex example had already disproved the original integer-root conjecture; no claim of global minimality at 30 is made. Three independent exact polynomial computations agree.

49. A marked-cycle proof of the trapezohedral domination conjecture

`graph-theory/trapezohedral-minimal-dominating-sets/article.pdf`
from `trapezohedral_domination_research.zip`

Proves the generating function labelled conjectural in OEIS A381190: for the n -trapezohedral graph, the number of dominating sets that are inclusion-minimal among all dominating sets and induce a connected subgraph is $a_n = 2n(2p_{n-2} + p_{n-3})$ with $p_m = p_{m-2} + p_{m-3}$ (Padovan). The proof classifies such sets by a cyclic arrangement of gaps of length 2 and 3 with a restricted marked edge. Adds the size distribution, strict log-concavity, an unbiased sampler, a Binet formula and a central limit theorem.

9 Enumerative combinatorics

50. A Twelve-Value Proof of the A181280 Formula

`enumerative-combinatorics/a181280-binary-matrix-formula/article.pdf`

from A181280_research_package.zip

The binary-matrix formula recorded as Conjecture 20 by Kauers and Koutschan (2023) and still marked conjectural in OEIS A181280: an explicit spectral recurrence bound and a finite certificate resting on twelve decisive counts, together with further consequences. STATUS.md records that general C-finiteness was already proved in Kauers’s December 2025 lecture slides.

51. Esthetic Numbers, Folded Paths, and Algebraic Diagonals

enumerative-combinatorics/esthetic-numbers-and-folded-paths/article.pdf

from esthetic_walks_research.zip

The diagonal conjecture $T(q, q - 2) = A182555(q - 2)$ in OEIS A377000, where $T(q, k)$ counts base- q positive integers of length k whose adjacent digits differ by one. Proves it exactly, proves the entry’s other two conjectures as retrieved, and derives formulas for every fixed-offset diagonal $q = k + d$.

52. Diagonal Hardinian Arrays: A Fixed-Parameter Asymptotic Formula

enumerative-combinatorics/hardinian-array-diagonal-asymptotics/article.pdf

from hardinian_arrays_research.zip

For every fixed $r \geq 0$, $H_r(n, n) \sim C_r 4^{rn} n^{-r(r-1)/2}$ with an explicit C_r in Gamma values. The formula confirms the constants tabulated by Dougherty-Bliss and Kauers (2024) but refutes their proposed restriction to powers of 2, 3 and π : $C_7 = 2^{1852}/(3^{45}\pi^3)$. Adds an exact finite determinant at fixed array size, an exact enumeration algorithm, and a limiting negative-binomial law for total deficits.

53. An explicit formula for lucky parking spots

enumerative-combinatorics/lucky-parking-spots/article.pdf

from lucky_parking_spots_research.zip

Proves Conjecture 13 of Butler, Hadaway, Lenius, Martens and Moats (JIS 29 (2026), Article 26.1.1): a closed form for $C(n, j)$, the number of parking functions in which *spot* j is lucky. Adds an exact reflection law, all fixed right-edge columns, a closed form for OEIS A374533, Poisson edge limits, uniform bulk convergence, polynomial reciprocity and asymptotic expansions.

54. Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

enumerative-combinatorics/adjacency-bounded-132-avoiders/article.pdf

from strict_growth_132.zip

For 132-avoiding permutations with every adjacent difference at most m and growth constant α_m : proves $\alpha_{m+1} > \alpha_m$ for all $m \geq 1$ — the remaining strict-inequality part of Nadler’s Conjecture 2 — and the refinement $4 - \alpha_m = (2 \log m + 4 \log \log m + O(1))/m$, with $2 \log \pi - 4 \log 2 \leq \liminf E_m \leq \limsup E_m \leq 2 \log \pi$. The all- m component ordering and convergence of E_m are explicitly not claimed.

55. Throwback Dynamics with Repeated Entries

enumerative-combinatorics/throwback-queue-dynamics/throwback_research.pdf

from throwback_research.zip

For the queue process that removes the head and reinserts it at the zero-based position equal to its weight: every token recurs if and only if at most m input tokens have weight at most m , for every $m \geq 1$, with an explicit description of the recurrent tokens when that fails. Adds exact cycle periods and frequencies, a coprimality criterion for perfect mixing, Catalan and prime-parking counts for recurrent cores, and the largest labelled period of a core of size r .

56. Reciprocity and Matrix Duality for Preorder Polytopes

enumerative-combinatorics/preorder-polytope-reciprocity/article.pdf

from preorder_reciprocity.zip

Proves Conjectures 7.2, 4.2 and 4.4 of Athanasiadis and Chapoton, *Polytopes and posets associated to preorders* (arXiv:2605.26916v1), and the ordinary $q = 1$ case of their Conjecture 4.9 (not its full q -refinement). Two mechanisms: Ehrhart–Macdonald reciprocity after subtracting the all-ones

vector, extending the double identity to independently weighted capacities; and duplication of both colour classes of the bipartite graph followed by Postnikov’s lattice-point formula, giving the conjectured matrix transpose.

57. Rationality and growth of stabilized permutation-weight series

[enumerative-combinatorics/stabilized-permutation-weights/article.pdf](#)

from [stabilized_permutation_weights.zip](#)

Studies the full stabilized generating functions of the permutation-weight q -Eulerian polynomials, beyond their initial connection with OEIS A256193: an all- d finite-spectrum theorem, rationality, a universal integer denominator, the exact radius of convergence and residue-class asymptotics. The two-descent sequence gets an explicit formula in powers of 3, powers of 2 and Fibonacci numbers.

10 Generating functions and asymptotics

10.1 Coefficient counts of binary products

Four reports on counting coefficients of $\prod_i(1+x^{G_i})$. The first three answer, independently, the universal-rationality part of Richard Stanley’s MathOverflow question 431075 (“A conjectured rational generating function”, 2022), which had no posted answer when checked on 19 September 2026.

58. When Rational Exponents Produce Non-D-Finite Coefficient Counts

[generating-functions-and-asymptotics/binary-product-coefficient-counts/](#)

[mixed-base-natural-boundary/article.pdf](#)

from [Stanley_Rationality_Counterexample.zip](#)

With $G(2j+1) = 2^j$ and $G(2j+2) = 3^j$, the exponent generating function is rational but the generating function of $\nu(N)$ — the number of coefficients equal to one in $P_N(x) = \prod_{i \leq N}(1+x^{G(i)})$ — is not D-finite and has a natural boundary; explicitly, $a(n) = \nu(2n) = 2^{\lfloor (n+1)(1-\log 2/\log 3) \rfloor}$ for $n \geq 2$. The proof assumes nothing about the algebraicity of the growth constant. In the family G_b the count series is rational exactly when b is a power of two.

59. A Mixed-Base Counterexample to Stanley’s Rationality Question

[generating-functions-and-asymptotics/binary-product-coefficient-counts/](#)

[mixed-base-rationality-classification/article.pdf](#)

from [stanley_mixed_base_counterexample.zip](#)

The same mixed-base construction, with $\nu(2n) = 2^{n-\lfloor (n+1)\alpha \rfloor}$ for $\alpha = \log 2/\log 3$ proved by interval coverage and a carry-gap count. Nonrationality is obtained by a finite-field argument, then strengthened to a natural boundary, non-D-finiteness and non-P-recursiveness. Classifies rationality in the family interleaving a^j and q^j with digit polynomial $1+y+\dots+y^{a-1}$: it holds exactly when $q^u = a^v$.

60. Rational exponents, nonrational coefficient counts

[generating-functions-and-asymptotics/binary-product-coefficient-counts/](#)

[recursive-exponent-construction/research.pdf](#)

from [stanley_counterexample.zip](#)

A different negative answer, from the recursion $a_0 = 2$, $a_1 = 1$, $a_n = a_{n-1} - 2a_{n-2}$, with $b_n = T_{n-1} + 2^n - 1 + a_n$, $T_n = T_{n-1} + b_n$, $G_{2n-1} = 2^{n-1}$ and $G_{2n} = b_n$: the counting series has a natural boundary on the unit circle, hence is neither rational, algebraic nor D-finite.

61. Odd coefficients in quasifibonacci products

[generating-functions-and-asymptotics/binary-product-coefficient-counts/](#)

[quasifibonacci-odd-coefficient-counts/article.pdf](#)

from [quasifibonacci_parity_research.zip](#)

Stanley’s Conjecture 6.3, with a seed-independent extension. For $k \geq 2$ and seeds each exceeding the sum of their predecessors, continued by the k -term recurrence, the number $h_k(n)$ of odd

coefficients of $\prod_{i \leq n} (1 + x^{w_i})$ has generating function $(1 + 2z^k)/(1 - 2z + 2z^k - 2z^{k+1})$, independently of the seeds. The proof builds periodic signs of block length $k + 1$ and block product -1 and shows every signed finite product is flat.

10.2 Asymptotics, records and discrepancy

62. The Exponential Scale of Gregory Sign Thresholds

[generating-functions-and-asymptotics/gregory-coefficient-sign-thresholds/gregory_sign_threshold.pdf](#)
 from [gregory_sign_threshold_research.zip](#)

For $(x/\log(1+x))^m = \sum_n G_n^{(m)} x^n$ and $\text{Gr}(m)$ the least N beyond which $(-1)^{n-1} G_n^{(m)} > 0$, proves $\log \text{Gr}(m) = m + 1 - \gamma - (\zeta(2) + \zeta(3))/m + C_2/m^2 + O(m^{-3})$ with C_2 explicit, plus a finite algorithm for every coefficient of the complete expansion; in particular $\text{Gr}(m)/e^m \rightarrow e^{1-\gamma}$ and $\text{Gr}(m)/3^m \rightarrow 0$. This replaces Conjecture 6.1 of Xu and Zhao (arXiv:2609.11072v1), whose printed lower bound already conflicts with its own reported $\text{Gr}(5) = 113$; that value is independently certified here.

63. Exact Records in the Square-Root-of-Three Walk

[generating-functions-and-asymptotics/sqrt-three-walk-records/article.pdf](#)
 from [sqrt3_record_recurrence.zip](#)

Proves the four-phase record recurrence for $S(N) = \sum_{n \leq N} (-1)^{\lfloor n\sqrt{3} \rfloor}$ in the corrected form on page 7 of Bruin and Fokkink, *On the records and zeros of a deterministic random walk* (arXiv:2503.11734v2), where it is experimental and attributed to Arias de Reyna and van de Lune. Includes minimality of every claimed first passage, a rational generating function, closed forms, sharp logarithmic extrema, and an extension to every even integer trace $D \geq 4$.

64. Sharp discrepancy for a family of binary substitutions

[generating-functions-and-asymptotics/binary-substitution-discrepancy/article.pdf](#)
 from [oeis_discrepancy_counterexamples.zip](#)

For the fixed word of $0 \mapsto 1, 1 \mapsto 101010$, the position sequences A284365 and A284366 both carry a conjectured upper position error below 2, and both conjectures are false: the first counterexamples are at ranks 2,977,771 and 8,933,313, with common error $(2977771\sqrt{21} - 13645857)/2 \approx 2.00487$ and sharp common upper bound $(6 + \sqrt{21})/5 \approx 2.11652$. Proves complete discrepancy limit intervals for every substitution $0 \mapsto 1, 1 \mapsto (10)^m$, with infinite counterexample families and exact recurrences for their indices.